Experiment No. 4

Name of experiment: Critical Resistance of D.C Generator

1. Object: *To determine the critical resistance and critical speed of DC generator*.

1- To determine the critical resistance of a separately excited DC generator.
2- To determine the critical resistance of a self-excited DC generator.
3- To determine the critical speed of a self-excited D.C generator.

- **Critical resistance of a separately excited DC generator.**

2. Theory:

The critical resistance is defined by the total resistance of the field circuit .If the critical resistance is larger than the field resistance the voltage can be build, but if it is smaller than the field resistance the voltage cannot be built .To determine the critical resistance we must draw the (no-load) curve at any speed, the tangent to this curve which is drawn from the original point is known as the critical resistance, i.e. the slope of this line gives the critical resistance.

3. Procedure:

1. Connect the circuit as shown in fig (1).
2. Take the reading when the field current equal to zero (If =0).
4. Increase the field current step-by-step and take the induced voltage readings.
5. Repeat these steps at speed to 750 rpm &1400 rpm.

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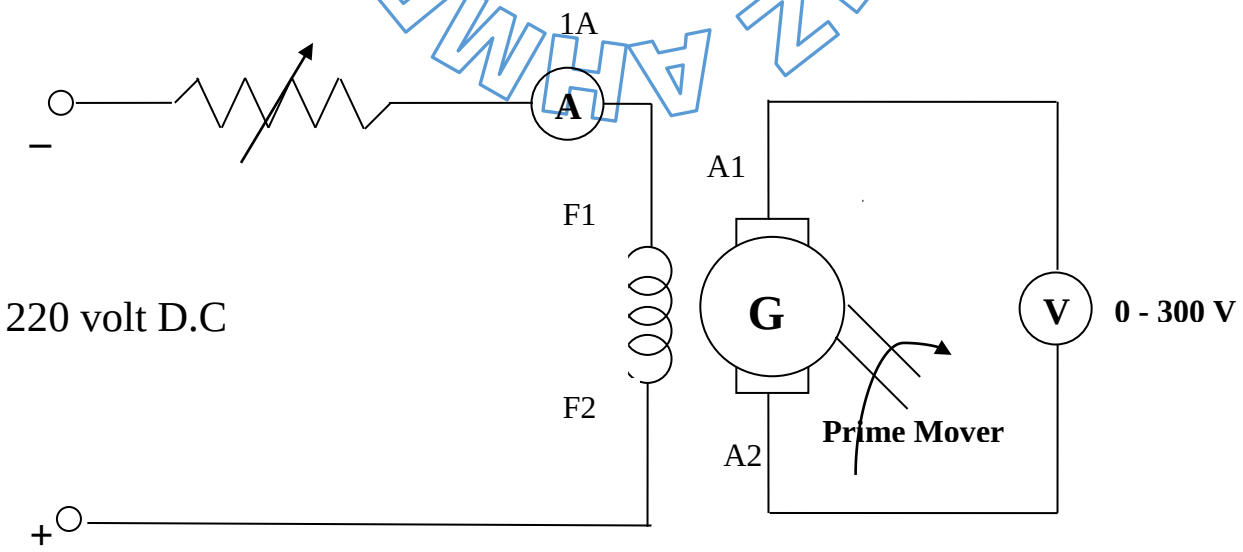


Fig (1)

N = 1400 rpm

If	0	0.1	0.2	0.3	0.4	0.5
Eo						

N = 750 rpm

If	0	0.1	0.2	0.3	0.4	0.5
Eo						

▪ Critical speed of a self-excited DC generator.

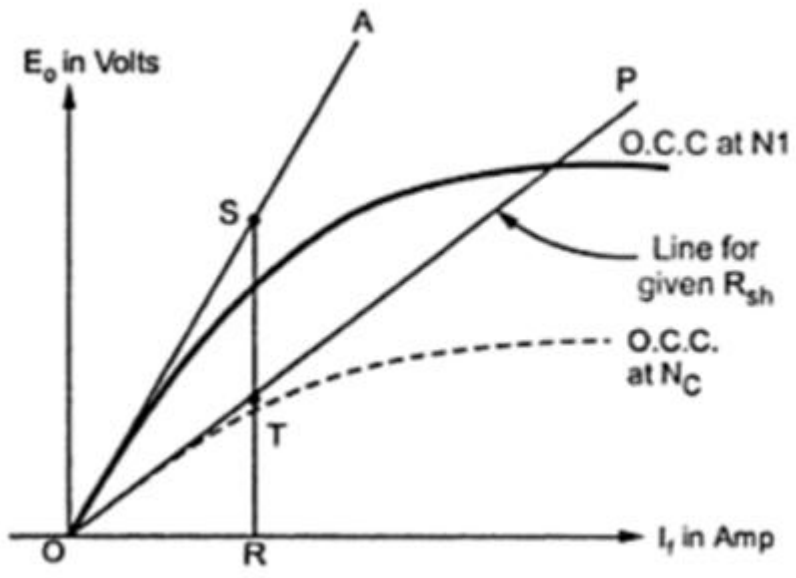
1. Object: To determine the critical speed of a self-excited D.C generator.

2. Theory: *Critical Speed* it is known that as speed changes, the open circuit characteristics also changes, similarly for different shunt field resistances, the corresponding lines are also different.
Note: The speed for which the given field resistance acts as critical resistance is called the critical speed, denoted as N_C. Thus if the line is drawn representing given R_{sh} then O.C.C. drawn for such a speed to which this line is tangential to the initial portion, is nothing but the critical speed N_C. Graphically critical speed can be obtained for given R_{sh}. The steps are:

1. Drawn O.C.C. for given speed N_1.
2. Draw a line tangential to this O.C.C. say OA.
3. Draw a line representing the given R_{sh} say OP.
4. Select any field current say point R.
5. Draw vertical line from R to intersect OA at S and OP at T.
6. Then the critical speed N_C is,

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While $\frac{RT}{RS} = \frac{N_C}{N_1}$ i.e $\boxed{N_C = N_1 \frac{RT}{RS}}$



$\boxed{N_C = N_1 \times \frac{RT}{RS}}$

3. Procedure:

1. Connect the circuit as shown in fig (1).
2. Fix the speed at 1400 rpm.
3. Record the armature voltage when the field current equal to zero.
4. Increase the field current and take the readings:

N = 1400 rpm

If	0	0.1	0.2	0.3	0.4	0.5	0.6
Eo							

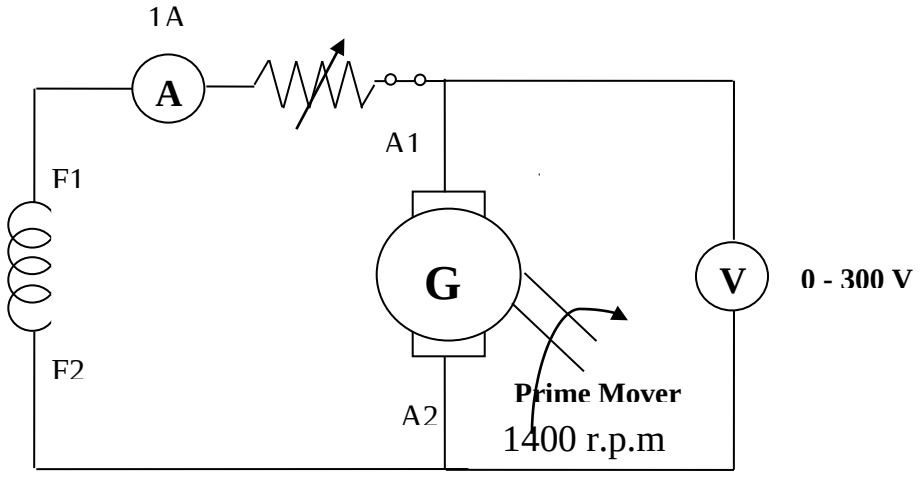


Fig (2)

4.1 Report (self-exited):

1. Plot the no-load curve.
2. From the step (1) determine the critical resistance.
3. What is the meaning of critical resistance?
4. List the factors that affect the voltage building in D.C generator.
5. Does the critical resistance affect the voltage building? Discuss briefly.
6. What is going to happen when you repeat this experiment on self D.C generator?

4.2 Report (self-exited)

1. Plot the no-load curve.
2. From the step (1) determine the critical resistance
3. From step (2) determine and calculate the critical speed.
4. How can you find the critical speed?
5. What is the effect of this speed on the generator in building E.M.F?
6. Discuss your results.

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